Conserve mechanical energy

antocciphysics | TEACHER KEY | CPA

Learning target: Track kinetic and potential energy when nonconservative work is negligible.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 4, Day 7; 3 course days later.*
+150 J; the object gains gravitational potential energy.

02. Explain a worked example (worked)
U 50 to 0, K 0 to 50; object-Earth system, negligible drag, height zero at arrival.

03. Look and explain (model)
5 J and 15 J.

04. Build the solution (practice)
20 J each.

05. Use the idea (reason)
No; include thermal/deformation/sound energy and surroundings as appropriate.

06. Choose a method (mixed) *Review: Unit 3, Day 12; 12 course days later.*
Both become 4 times as large; p=mv and average force magnitude=change in momentum/time.

07. Exit check (exit)
18 J.

08. STRETCH: Apply it another way (stretch)
Same final speed for equal initial speeds; path length, time and acceleration histories can differ.

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01. Bring back an earlier idea (retrieve) *Review: Unit 4, Day 7; 3 course days later.*
$\Delta U = -150$ J; gravity does +150 J work.

02. Explain a worked example (worked)
U 50 to 0, K 0 to 50; object-Earth system, negligible drag, height zero at arrival.

03. Represent and interpret (model)
5 J and 15 J.

04. Solve and justify (practice)
Initial E $100+16=116$ J; final U 20, K 96; $v=\sqrt{96}=9.80$ m/s.

05. Analyze the reasoning (reason)
No; include thermal/deformation/sound energy and surroundings as appropriate.

06. Choose a method (mixed) *Review: Unit 3, Day 12; 12 course days later.*
A has 1/4 of B's speed and kinetic energy: $v=p/m$, $K=p^2/(2\,m)$.

07. Exit check (exit)
$v^2/(2\,g)=5$ m.

08. STRETCH: Apply it another way (stretch)
Initial K=16 J; 3 m needs 60 J, so no. Maximum rise $v^2/(2\,g)=0.8$ m.